

UrbanFix Training Start Guide (Final, Download-Only Flow)

This is the exact sequence to follow if you do not want to build your own dataset now.

A) Create Drive folders first

Create these folders in Google Drive:

```
MyDrive/UrbanFix_AI/  
MyDrive/UrbanFix_AI/datasets/  
MyDrive/UrbanFix_AI/datasets/trash_yolo/  
MyDrive/UrbanFix_AI/datasets/lighting_yolo/  
MyDrive/UrbanFix_AI/datasets/parks_yolo/  
MyDrive/UrbanFix_AI/datasets/water_cls/  
MyDrive/UrbanFix_AI/models/  
MyDrive/UrbanFix_AI/runs/trash/  
MyDrive/UrbanFix_AI/runs/lighting/  
MyDrive/UrbanFix_AI/runs/parks/
```

B) Download datasets (no custom labeling required to start)

1) Trash

- Source: TACO / Roboflow garbage dataset exported in YOLOv8 format.
- Required final location:
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/trash_yolo/images/train`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/trash_yolo/images/val`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/trash_yolo/labels/train`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/trash_yolo/labels/val`

2) Lighting

- Source: street-light/pole dataset exported in YOLOv8 format.
- Required final location:
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/lighting_yolo/images/train`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/lighting_yolo/images/val`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/lighting_yolo/labels/train`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/lighting_yolo/labels/val`

3) Parks

- Source: park/public asset dataset exported in YOLOv8 format.
- Required final location:
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/parks_yolo/images/train`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/parks_yolo/images/val`
 - `/content/drive/MyDrive/UrbanFix_AI/datasets/parks_yolo/labels/train`

- `/content/drive/MyDrive/UrbanFix_AI/datasets/parks_yolo/labels/val`

4) Water (classification)

- Source: Roadway Flooding image dataset (Kaggle) + any flood/non-flood open images.
- Required final location:

```
/content/drive/MyDrive/UrbanFix_AI/datasets/water_cls/  
train/flooded/*  
train/non_flooded/*  
val/flooded/*  
val/non_flooded/*
```

C) Put dataset.yaml templates

Copy these files and rename/place as below:

- `colab/dataset_trash.yaml` -> `.../datasets/trash_yolo/dataset.yaml`
- `colab/dataset_lighting.yaml` -> `.../datasets/lighting_yolo/dataset.yaml`
- `colab/dataset_parks.yaml` -> `.../datasets/parks_yolo/dataset.yaml`

Important: if your class names differ, edit `dataset.yaml` names to match your downloaded labels.

D) Run notebooks/scripts in this exact order

1. `UrbanFix_Trash_Train.py`
2. `UrbanFix_Water_Train.py`
3. `UrbanFix_Lighting_Train.py`
4. `UrbanFix_Parks_Train.py`

E) Expected outputs

- Trash: `MyDrive/UrbanFix_AI/models/trash_best.pt`
- Water: `MyDrive/UrbanFix_AI/models/water_flood_siglip/` (folder)
- Lighting: `MyDrive/UrbanFix_AI/models/lighting_best.pt`
- Parks: `MyDrive/UrbanFix_AI/models/parks_best.pt`

F) Minimum acceptance metrics

- Trash mAP50 ≥ 0.72
- Water weighted F1 ≥ 0.90
- Lighting mAP50 ≥ 0.65
- Parks mAP50 ≥ 0.60